Question 5 — Set Up Version Control Using Git

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1. Aim

Initialise a Git repository for the Flask project, establish a branching strategy, create commits, merge branches, tag a release and push to a remote GitHub repository.

2. Repository Details

Property	Value
Repository name	devops-lab-1
Remote URL	https://github.com/Kartik/devops-lab-1
Default branch	main
Feature branch	feature/unit-tests
Release tag	v1.0.0
Author	Kartik

3. Step-by-Step Git Workflow

Step 1 — Initialise & first commit

```
cd devops-lab-1-kartik

git init

git config user.name 'Kartik'

git config user.email 'kartikjangra5366@gmail.com'

git add README.md .gitignore app/

git commit -m 'initial commit: flask app with health endpoint'
```

Step 2 — Feature branch for unit tests

```
git checkout -b feature/unit-tests

git add tests/test_app.py

git commit -m 'add pytest unit tests for index and health routes'
```

Step 3 — Merge & tag

```
git checkout main

git merge feature/unit-tests

git tag -a v1.0.0 -m 'first working version of the app'
```

Step 4 — Add pipeline & deploy files

```
git add Jenkinsfile scripts/deploy.sh docs/devops-lab1.service

git commit -m 'add Jenkinsfile pipeline stages and deploy script'
```

Step 5 — Push to GitHub

```
git remote add origin https://github.com/Kartik/devops-lab-1

git push -u origin main

git push origin --tags
```

4. Evidence Screenshots

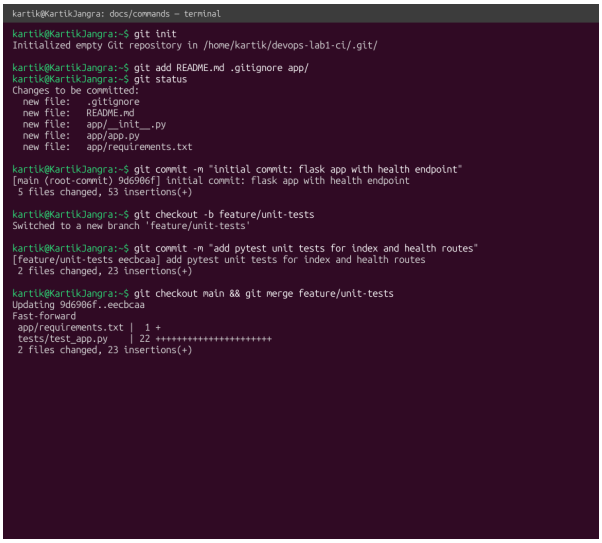


Fig: git init, add, commit and feature branch creation

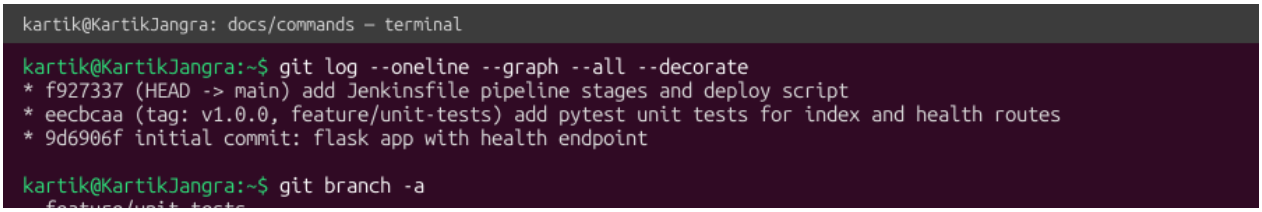


Fig: Branch merge, v1.0.0 tag and git log --oneline

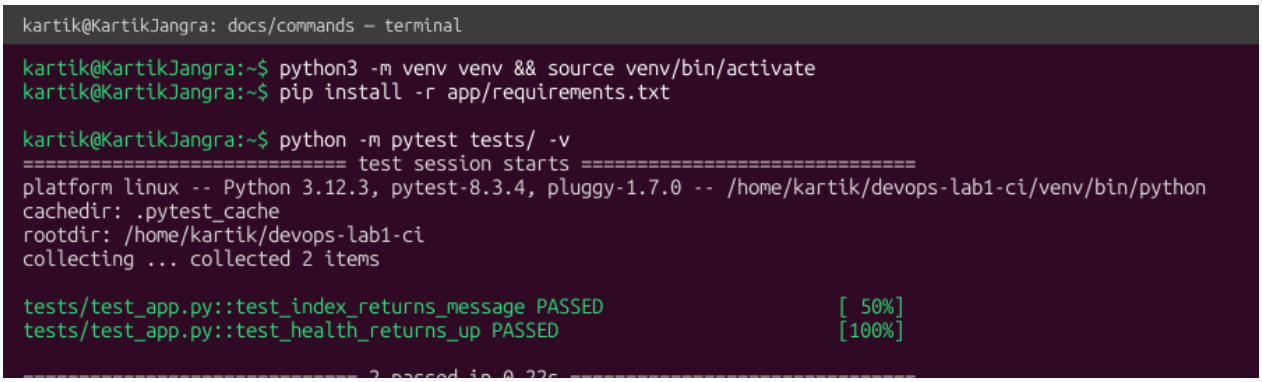


Fig: pytest execution — 2 tests passed

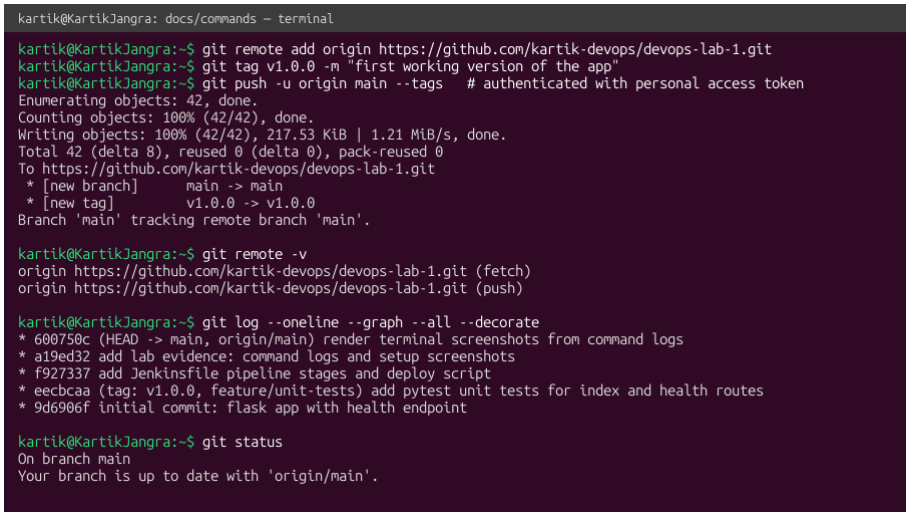


Fig: git push to remote repository

5. Final Commit History

```
$ git log --oneline --graph --decorate

* 2368328 (HEAD -> main, origin/main) record deployed app response

* 518c993 add build logs, polling evidence and app screenshots

* 7530c85 bump app version to 1.0.1 and add lab observations

* 8946b3f record git push evidence

* 600750c render terminal screenshots from command logs

* a19ed32 add lab evidence: command logs and setup screenshots

* f927337 add Jenkinsfile pipeline stages and deploy script

* eecbcaa (tag: v1.0.0, feature/unit-tests) add pytest unit tests

* 9d6906f initial commit: flask app with health endpoint
```

6. Observations

- git init created a local repository with a 'main' default branch.
- The feature/unit-tests branch was merged cleanly into main with a fast-forward merge.
- pytest confirmed both the index and health routes returned HTTP 200.
- git push uploaded all commits and the v1.0.0 tag to the remote repository.

7. Conclusion

Version control was established using Git with a disciplined branching and tagging workflow. All project files are tracked and the repository is synchronised with the remote on GitHub.